

Lemma 1 (Pairwise truncation bound). *Let $0 < \varepsilon \leq 1$ and $0 < \eta \leq \varepsilon^2/16$. For any a priori tour T , we have*

$$\text{cost}_\eta^{\text{loc}}(T) \leq \text{cost}(T) \leq (1 + \varepsilon) \text{cost}_\eta^{\text{loc}}(T).$$

Proof. The first inequality is immediate because the truncated cost omits only nonnegative terms. We prove the second inequality.

For $0 \leq i < j \leq n+1$, let

$$w_{ij} := p_i p_j Q(i, j),$$

and write

$$C := \text{cost}(T), \quad C_\eta := \text{cost}_\eta^{\text{loc}}(T), \quad D_\eta := C - C_\eta.$$

Thus, D_η is the total contribution of the discarded pairs, namely those pairs (i, j) for which $Q(i, j) \leq \eta$.

Fix a discarded pair (i, j) . Independently sample the activation states of the vertices strictly between v_i and v_j , conditional on at least one of them being active. This conditioning event has probability

$$1 - Q(i, j) \geq 1 - \eta.$$

Together with v_i and v_j , the sampled active vertices form a path in the order in which they occur in T . Denote its edge set by P_{ij} . By the triangle inequality,

$$d_{ij} \leq \sum_{(a,b) \in P_{ij}} d_{ab}.$$

We use this inequality to charge $w_{ij}d_{ij}$ to the edges of the random path P_{ij} .

Fix a target pair (a, b) with $w_{ab} > 0$, and let Γ_{ab} be the expected coefficient charged to d_{ab} . A source pair (i, j) can charge (a, b) only if $i \leq a < b \leq j$. If $(i, j) \neq (a, b)$, then, before conditioning, the probability that a and b are consecutive on the sampled path is

$$Q(a, b) p_a^{1[i < a]} p_b^{1[b < j]}.$$

Conditioning increases this probability by at most a factor $1/(1 - \eta)$. After dividing by w_{ab} and separating the three possible strict-containment patterns, we obtain

$$\begin{aligned} \frac{\Gamma_{ab}}{w_{ab}} \leq \frac{1}{1 - \eta} \Bigg(& \sum_{\substack{j > b \\ Q(a, j) \leq \eta}} p_j Q(a, j) + \sum_{\substack{i < a \\ Q(i, b) \leq \eta}} p_i Q(i, b) \\ & + \sum_{\substack{i < a, j > b \\ Q(i, j) \leq \eta}} p_i p_j Q(i, j) \Bigg). \end{aligned} \quad (1)$$

The first two sums in (1) are each at most η . Indeed, $Q(a, j)$ is nonincreasing in j , so the indices satisfying $Q(a, j) \leq \eta$ form a suffix, and telescoping gives

$$\sum_{\substack{j > b \\ Q(a, j) \leq \eta}} p_j Q(a, j) \leq \eta.$$

The second sum is bounded symmetrically.

It remains to bound the third sum in (1). Set

$$H := (1 - p_a)Q(a, b)(1 - p_b), \quad x_i := Q(i, a), \quad y_j := Q(b, j).$$

For $i < a < b < j$, we have $Q(i, j) = Hx_i y_j$. Moreover, telescoping gives

$$\sum_{i < a} p_i x_i = 1, \quad \sum_{j > b} p_j y_j = 1.$$

If either index set is empty, the third sum is zero. If $H \leq \eta$, it is at most H and hence at most η . Suppose, therefore, that $H > \eta$, and put $t := \eta/H$. For every $z > 0$, telescoping on the right gives

$$\sum_{\substack{j > b \\ y_j \leq z}} p_j y_j \leq \min\{1, z\}.$$

Terms with $x_i = 0$ vanish. Consequently,

$$\begin{aligned} \sum_{\substack{i < a, j > b \\ Q(i, j) \leq \eta}} p_i p_j Q(i, j) &= H \sum_{\substack{i < a \\ x_i > 0}} p_i x_i \sum_{\substack{j > b \\ y_j \leq t/x_i}} p_j y_j \\ &\leq H \sum_{i < a} p_i \min\{x_i, t\}. \end{aligned}$$

We use the following elementary survival-product estimate:

$$\sum_{i < a} p_i \min\{Q(i, a), t\} \leq t \left(1 + \ln \frac{1}{t}\right) \quad (0 < t < 1). \quad (2)$$

To verify it, recall that $x_i = Q(i, a)$ is nondecreasing in i and that

$$x_{i-1} = (1 - p_i)x_i.$$

If $x_0 > t$, then

$$\sum_{i < a} p_i \min\{x_i, t\} = t \sum_{i < a} p_i \leq t \left(1 + \ln \frac{1}{x_0}\right) \leq t \left(1 + \ln \frac{1}{t}\right).$$

Otherwise, let ℓ be the largest index for which $x_\ell \leq t$, and put $x := x_{\ell+1}$ and $z := x_\ell$. Since $z = (1 - p_{\ell+1})x$, telescoping and the inequality $u \leq -\ln(1 - u)$ give

$$\begin{aligned} \sum_{i < a} p_i \min\{x_i, t\} &\leq z + t \left(1 - \frac{z}{x} + \ln \frac{1}{x}\right) \\ &\leq t \left(2 - \frac{t}{x} + \ln \frac{1}{x}\right) \\ &\leq t \left(1 + \ln \frac{1}{t}\right). \end{aligned}$$

The last inequality is equivalent to $1 - t/x + \ln(t/x) \leq 0$, which follows from $\ln u \leq u - 1$. This proves (2).

Applying (2), the third sum in (1) is at most

$$Ht \left(1 + \ln \frac{1}{t}\right) = \eta \left(1 + \ln \frac{H}{\eta}\right) \leq \eta \left(1 + \ln \frac{1}{\eta}\right).$$

It follows from (1) that every target pair has congestion at most

$$\beta(\eta) := \frac{\eta(3 + \ln(1/\eta))}{1 - \eta}.$$

Pairs with $w_{ab} = 0$ receive no charge, so summing over all target pairs gives

$$D_\eta \leq \beta(\eta)C.$$

Therefore,

$$C_\eta \geq (1 - \beta(\eta))C. \tag{3}$$

The function β is increasing on $(0, 1)$. Since $\eta \leq \varepsilon^2/16 \leq 1/16$, we have

$$\begin{aligned} \beta(\eta) &\leq \beta\left(\frac{\varepsilon^2}{16}\right) \\ &\leq \frac{\varepsilon^2}{15} \left(3 + \ln 16 + 2 \ln \frac{1}{\varepsilon}\right) \\ &\leq \frac{\varepsilon^2}{15} \left(4 + \frac{2}{\varepsilon}\right) \\ &\leq \frac{2\varepsilon}{5} \leq \frac{\varepsilon}{1 + \varepsilon}. \end{aligned}$$

Here we used $\ln(1/\varepsilon) \leq 1/\varepsilon - 1$, $\ln 16 < 3$, and $0 < \varepsilon \leq 1$. Substituting this bound into (3) gives

$$C_\eta \geq \frac{C}{1 + \varepsilon},$$

which is the desired second inequality. □